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THEORIA RESIDUORUM BIQUADRATICORUM
COMMENTATIO SECUNDA

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SOCIETATI REGIAE TRADITA 1831. APR. 15

Commentationes Societatis Regiae Scientiarum Gottingensis recentiores, Vol. VII.
Gottingae MDCCCXXXII.

24.

In commentatione prima ea, quae ad classificationem biquadraticam numeri $+2$ requiruntur, complete absoluta sunt. Dum scilicet omnes numeros per modulum p (qui supponitur esse numerus primus formae $4n + 1$) non divisibiles inter quatuor complexus A, B, C, D distributos concipimus, prout singuli ad potestatem exponentis $\frac{1}{4}(p-1)$ evecti congrui fiunt secundum modulum p ipsi $+1, +f, -1, -f$, denotante f radicem alterutram congruentiae $ff \equiv -1 \pmod{p}$: invenimus, diiudicationem, cuinam complexui adnumerandus sit numerus $+2$, pendere a discriptione numeri p in duo quadrata, ita quidem, ut si statuatur $p = aa + bb$, denotante aa quadratum impar, bb quadratum par, si porro signa ipsorum a, b ita accepta supponantur, ut habeatur $a \equiv 1 \pmod{4}$, $b \equiv af \pmod{p}$, numerus $+2$ ad complexum A, B, C, D pertinere debeat prout $\frac{1}{2}b$ sit formae $4n, 4n + 1, 4n + 2, 4n + 3$ resp.

Sponte quoque hinc demanat regula classificationi numeri -2 inserviens. Scilicet quum -1 pertineat ad classem A pro valore pari ipsius $\frac{1}{2}b$, ad classem C vero pro impari, pertinebit, per theorema art. 7, numerus -2 ad classem A, B, C, D , prout $\frac{1}{2}b$ est formae $4n, 4n + 3, 4n + 2, 4n + 1$ resp. Haec theoremata etiam sequenti modo exprimi possunt:

Pertinet ad complexum	$+2$	-2	
A	$b \equiv 0$	$b \equiv 0$	
B	$b \equiv 2a$	$b \equiv 6a$	$(\text{mod } 8).$
C	$b \equiv 4a$	$b \equiv 4a$	
D	$b \equiv 6a$	$b \equiv 2a$	

Facile intelligitur, theoremata sic enunciata haud amplius pendere a conditione $a \equiv 1 \pmod{4}$, sed etiamnum valere, si fuerit $a \equiv 3 \pmod{4}$, dummodo conditio altera, $af \equiv b \pmod{p}$, conservetur. Aequae facile perspicitur, summam horum theorematum eleganter contrahi posse in formulam unicam, puta: si a et b positive accipiuntur, semper fit

$$b^{\frac{1}{2}ab} \equiv a^{\frac{1}{2}ab} 2^{\frac{1}{4}(p-1)} \pmod{p}.$$

25.

Videamus nunc, quatenus inductio classificationem numeri 3 indigitet. Tabula art. 11 ulterius continuata (semper adoptata radice primitiva minima), monstrat, +3 pertinere ad complexum

A pro				B pro			C pro			D pro	
p	a	b	p	a	b	p	a	b	p	a	b
13	-3	+2	17	+1	-4	37	+1	-6	5	+1	+2
109	-3	+10	29	+5	+2	61	+5	-6	41	+5	-4
181	+9	+10	53	-7	+2	73	-3	-8	149	-7	+10
193	-7	-12	89	+5	-8	97	+9	+4	173	+13	+2
229	-15	+2	101	+1	+10	157	-11	-6			
277	+9	+14	113	-7	-8	241	-15	-4			
			137	-11	-4						
			197	+1	-14						
			233	+13	+8						
			257	+1	-16						
			269	+13	+10						
			281	+5	+16						
			293	+17	+2						

Primo saltem aspectu nexum simplicem inter valores numerorum a, b , quibus idem complexus respondet, non animadvertimus. At si perpendimus, diiudicationem similem in theoria residuorum quadraticorum per regulam simpliciore absolvere respectu numeri -3 , quam respectu numeri $+3$, spes affulget successus aequae secundi in theoria residuorum biquadraticorum. Invenimus autem, -3 pertinere ad complexum

A pro				B pro			C pro			D pro	
p	a	b	p	a	b	p	a	b	p	a	b
37	+1	-6	5	+1	+2	13	-3	+2	29	+5	+2
61	+5	-6	17	+1	-4	73	-3	-8	41	+5	-4
157	-11	-6	89	+5	-8	97	+9	+4	53	-7	+2
193	-7	-12	113	-7	-8	109	-3	+10	101	+1	+10
			137	-11	-4	181	+9	+10	197	+1	-14
			149	-7	+10	229	-15	+2	269	+13	+10
			173	+13	+2	241	-15	-4	293	+17	+2
			233	+13	+8	277	+9	+14			
			257	+1	-16						
			281	+5	+16						

ubi lex inductionis sponte se offert. Scilicet pertinet -3 ad complexum

$$\begin{array}{l|l}
 A & b \equiv 0 \pmod{3} \\
 B & a + b \equiv 0 \pmod{3}, \text{ sive } b \equiv 2a \pmod{3} \\
 C & a \equiv 0 \pmod{3} \\
 D & a - b \equiv 0 \pmod{3}, \text{ sive } b \equiv a \pmod{3}.
 \end{array}$$

26.

Numerum $+5$ adscribendum invenimus complexui

A	$p = 101, 109, 149, 181, 269$
B	$p = 13, 17, 73, 97, 157, 193, 197, 233, 277, 293$
C	$p = 29, 41, 61, 89, 229, 241, 281$
D	$p = 37, 53, 113, 137, 173, 257.$

In considerationem vocatis valoribus numerorum a, b singulis p respondentibus, lex hic aequae facile, ut pro classificatione numeri -3 , prehenditur. Scilicet incidimus in complexum

A	$b \equiv 0 \pmod{5}$
B	$b \equiv a \pmod{5}$
C	$a \equiv 0 \pmod{5}$
D	$b \equiv 4a \pmod{5}.$

Manifestum est, has regulas complecti casus omnes, quum pro $b \equiv 2a$, vel $b \equiv 3a \pmod{5}$, fieret $aa + bb \equiv 0$, Q.E.A., quum per hypothesin p sit numerus primus a 5 diversus.

27.

Perinde inductio ad numeros $-7, -11, +13, +17, -19, -23$ applicata satisque producta sequentes regulas indigitat:

Pro numero -7	
A	$a \equiv 0, \text{ vel } b \equiv 0 \pmod{7}$
B	$b \equiv 4a, \text{ vel } 5a$
C	$b \equiv a, \text{ vel } 6a$
D	$b \equiv 2a, \text{ vel } 3a$

Pro numero -11	
A	$b \equiv 0, 5a, \text{ vel } 6a \pmod{11}$
B	$b \equiv a, 3a, \text{ vel } 4a$
C	$a \equiv 0, \text{ vel } b \equiv 2a, \text{ vel } 9a$
D	$b \equiv 7a, 8a, \text{ vel } 10a$

Pro numero $+13$	
A	$b \equiv 0, 4a, 9a \pmod{13}$
B	$b \equiv 6a, 11a, 12a$
C	$a \equiv 0; b \equiv 3a, 10a$
D	$b \equiv a, 2a, 7a$

Pro numero $+17$	
A	$a \equiv 0; b \equiv 0, a, 16a \pmod{17}$
B	$b \equiv 2a, 6a, 8a, 14a$
C	$b \equiv 5a, 7a, 10a, 12a$
D	$b \equiv 3a, 9a, 11a, 15a$

Pro numero -19	
A	$b \equiv 0, 2a, 5a, 14a, 17a \pmod{19}$
B	$b \equiv 3a, 7a, 11a, 13a, 18a$
C	$a \equiv 0; b \equiv 4a, 9a, 10a, 15a$
D	$b \equiv a, 6a, 8a, 12a, 16a$

Pro numero -23	
A	$a \equiv 0; b \equiv 0, 7a, 10a, 13a, 16a \pmod{23}$
B	$b \equiv 2a, 3a, 4a, 11a, 15a, 17a$
C	$b \equiv a, 5a, 9a, 14a, 18a, 22a$
D	$b \equiv 6a, 8a, 12a, 19a, 20a, 21a$

28.

Theoremata specialia hoc modo per inductionem eruta confirmari inveniuntur, quousque haec continuetur, formamque criteriorum pulcherrimam manifestant. Si vero inter se conferuntur, ut conclusiones generales inde petantur, primo statim aspectu se offerunt observationes sequentes.

Criteria diiudicationis, ad quemnam complexum referendus sit numerus primus $\pm q$ (sumendo signum superius vel inferius, prout q est formae $4n+1$ vel $4n+3$), pendent a formis numerorum a, b inter se collatorum respectu moduli q . Scilicet:

I. quoties $a \equiv 0 \pmod{q}$, $\pm q$ pertinet ad complexum determinatum, qui est A pro $q = 7, 17, 23$, nec non C pro $q = 3, 11, 13, 19$, unde coniectura oritur, casum priorem generaliter valere, quoties q sit formae $8n \pm 1$, posteriorem vero, quoties q sit formae $8n \pm 3$. Ceterum complexus B et D iam absque inductione excluduntur pro valore ipsius a per q divisibili, ubi fit $p \equiv bb \pmod{q}$, i.e. ubi p est residuum quadraticum ipsius q , unde per theorema fundamentale $\pm q$ esse debet residuum quadraticum ipsius p .

II. Quoties autem a per q non est divisibilis, criterium pendet a valore expressionis $\frac{b}{a} \pmod{q}$. Admittit quidem haec expressio q valores diversos, puta $0, 1, 2, 3, \dots, q-1$; sed quoties q est formae $4n+1$, excludendi sunt bini valores expressionis $\sqrt{-1} \pmod{q}$, qui manifesto nequeunt esse valores expressionis $\frac{b}{a} \pmod{q}$, quum $p = aa + bb$ semper supponatur esse numerus primus a q diversus. Quapropter multitudo valorum admissibilium expressionis $\frac{b}{a} \pmod{q}$ est $q-2$ pro $q \equiv 1 \pmod{4}$, dum manet q pro $q \equiv 3 \pmod{4}$.

Iam hi valores in quaternas classes distribuuntur, puta, ut quidam, indefinite per α denotandi, respondeant complexui A ; alii per β denotandi complexui B ; alii γ complexui C ; denique reliqui δ complexui D , ita scilicet, ut $\pm q$ complexui A, B, C, D adscribendus sit, prout habeatur $b \equiv \alpha a, b \equiv \beta a, b \equiv \gamma a, b \equiv \delta a \pmod{q}$. At lex huius distributionis abstrusior videtur, etiamsi quaedam generalia prompte animadvertantur. Multitudo in ternis classibus eadem reperitur, puta $= \frac{1}{4}(q-1)$ vel $\frac{1}{4}(q+1)$, dum in una, et quidem in eadem, quae respondet complexui cum criterio $a \equiv 0$, unitate minor est, ita ut multitudo omnium criteriorum diversorum respectu singulorum complexuum fiat eadem, puta $= \frac{1}{4}(q-1)$ vel $\frac{1}{4}(q+1)$.

Porro animadvertimus, 0 semper in prima classe (inter α) reperiri nec non complementa numerorum $\alpha, \beta, \gamma, \delta$ ad q , puta $q-\alpha, q-\beta, q-\gamma, q-\delta$ resp. in classe prima, quarta, tertia, secunda. Denique valores expressionum $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}, \frac{1}{\delta} \pmod{q}$ pertinere videmus ad classem primam, quartam, tertiam, secundam, quoties criterium $a \equiv 0$ respondet complexui A ; ad classem tertiam, secundam, primam, quartam resp. autem, quoties criterium $a \equiv 0$ refertur ad com-

plexum C . Sed ad haec fere limitantur, quae per inductionem assequi licet, nisi audacius ea, quae infra e fontibus genuinis haurientur, anticipare nobis arrogemus.

29.

Antequam ulterius progrediamur, observare convenit, criteria pro numeris primis (positive sumtis, si sunt formae $4n + 1$, negative, si formae $4n + 3$) sufficere ad diiudicationem pro omnibus reliquis numeris, si modo theorema art. 7, atque criteria pro -1 et ± 2 in subsidium vocentur. Ita e.g. si desiderantur criteria pro numero $+3$, criteria in art. 25 prolata, quae referuntur ad -3 , etiamnum pro $+3$ valebunt, quoties $\frac{1}{2}b$ est numerus par; contra complexus A, B, C, D cum complexibus C, D, A, B permutandi erunt, quoties $\frac{1}{2}b$ est impar, unde sequuntur praecepta haecce:

+3 pertinet	
A	$b \equiv 0 \pmod{12}$; vel simul $a \equiv 0 \pmod{3}$, $b \equiv 2 \pmod{4}$
B	$b \equiv 8a$, vel $10a \pmod{12}$
C	$b \equiv 6a \pmod{12}$; vel simul $a \equiv 0 \pmod{3}$, $b \equiv 0 \pmod{4}$
D	$b \equiv 2a$, vel $4a \pmod{12}$.

Perinde criteria pro ± 6 petuntur e combinatione criteriorum pro ∓ 2 et -3 ; scilicet

+6 pertinet	
A	$b \equiv 0, 2a, 22a \pmod{24}$; vel simul $a \equiv 0 \pmod{3}$, $b \equiv 4a \pmod{8}$
B	$b \equiv 4a, 6a, 8a \pmod{24}$; vel simul $a \equiv 0 \pmod{3}$, $b \equiv 2a \pmod{8}$
C	$b \equiv 10a, 12a, 14a \pmod{24}$; vel simul $a \equiv 0 \pmod{3}$, $b \equiv 0 \pmod{8}$
D	$b \equiv 16a, 18a, 20a \pmod{24}$; vel simul $a \equiv 0 \pmod{3}$, $b \equiv 6a \pmod{8}$

et

-6 vero	
A	$b \equiv 0, 10a, 14a \pmod{24}$; vel simul $a \equiv 0 \pmod{3}$, $b \equiv 4a \pmod{8}$
B	$b \equiv 4a, 8a, 18a \pmod{24}$; vel simul $a \equiv 0 \pmod{3}$, $b \equiv 6a \pmod{8}$
C	$b \equiv 2a, 12a, 22a \pmod{24}$; vel simul $a \equiv 0 \pmod{3}$, $b \equiv 0 \pmod{8}$
D	$b \equiv 6a, 16a, 20a \pmod{24}$; vel simul $a \equiv 0 \pmod{3}$, $b \equiv 2a \pmod{8}$.

Simili modo criteria pro numero $+21$ concinnabuntur e criteriis pro -3 et -7 ; criteria pro -105 e criteriis pro $-1, -3, +5, -7$, etc.